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# A survey on deep reinforcement learning for personalized treatment featuring challenges and a theoretical framework

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**Abstract**

Personalized treatment is a paradigm shift in the medical field that provides individualized treatment strategies based on comprehensive patient data including habitat, genomic, and medical profile, etc. Integrating this into the conventional clinical process involves collaborative strategic planning with stakeholders from diverse sectors. The scope of this survey is limited to studies that apply reinforcement learning, deep reinforcement learning, and multi-objective deep reinforcement learning to sequential decision making in personalized treatment. The review addresses these main questions: Why are reinforcement learning and its variants suitable for personalized treatment?How are these algorithms applied in clinical settings of various domains? How are states, actions and rewards defined? What are the requirements, challenges at every stage of model development for deploying these models in safety-critical healthcare applications? Early studies primarily focused on reinforcement learning with single objective, while more works have increasingly adopted deep variants to manage high-dimensional state-action spaces, and a sparse literature exists with multi-objective approach to address conflicting treatment options. Most existing research is concentrated on using publicly available MIMIC- III / IV dataset in ICU settings, oncology related resources including, The Cancer Genome Atlas, Genomics of Drug Sensitivity in Cancer, The Catalogue of Somatic Mutations In Cancer. To the best of our knowledge, this is the first survey to emphasize: (i) the need for multi-objective optimization with sequential personalized treatment planning, (ii) the integration of non-clinical objectives such as patient preferences, hospital infrastructure, insurance coverage, etc., and stake-holders into decision framework, (iii) the theoretical and conceptual model of a multi-modal multi-objective deep reinforcement learning aiming at optimizing the treatment regimes using a newly introduced time-weighted reward integration block to balance multiple, potentially conflicting clinical as well as non-clinical objectives.

**Keywords** Affordable drugs, Clinical decision support, Deep reinforcement learning, Multi-objective optimization, Patient care, Personalized treatment, Precision medicine, Reinforcement learning

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## 1 Introduction

Precision medicine is a sequential medical treatment model that focuses on providing a tailored and more accurate diagnosis with a specific treatment strategy to a subgroup of the population who exhibit similar responses to a particular treatment [1]. Each individual within a species exhibits unique differences in both anatomical structure and biochemical activity. For this reason, Pharmacokinetics that defines the rate of Absorption, Distribution, Metabolism, and Excretion (ADME) of a drug is unique for each individual [2]. The factors that lead to similarity among the members of the subgroup or the differences between the different subgroups could be due to genetics, age, gender, race, ethnicity, environmental factors, socioeconomic status, etc. Personalized treatment strategies also referred to as dynamic treatment strategies have been experimented with for the past twenty years using statistical modeling approaches like parametric, semi-parametric, nonparametric, likelihood-based, etc. [3] using observational data. In traditional medical practice, treatment strategies are often standardized across the entire patient population, offering the same interventions regardless of individual variability. In contrast, personalized medicine is a data-driven decision support model that seeks to tailor treatment plans to the unique characteristics of each patient. These individualized approaches are designed to optimize both immediate clinical outcomes and long-term health. However, identifying relevant patient subgroups and determining the most effective treatment strategy often requires significant clinical expertise and a comprehensive review of the patient's medical history, diagnostics, other contextual data and integration of multi-omic data. This process is time-intensive and subject to variability in the clinician's judgment. Personalized treatment is still in an infancy stage, and a few innovations in this field are genetic profile specific therapies, personalized disease prevention and surveillance, etc. [2, 4, 5]. Personalized treatments are explored in various medical fields such as oncology [6–8], diabetes management [9], self-management [10], etc.

Personalized treatment can be formulated as a multi-objective sequential-decision problem that captures the current state of the patient from a high-dimensional clinical and non-clinical data, and provides a dynamic treatment strategy that balances a set of potentially conflicting objectives corresponding to different treatment options. The dynamic treatment strategy adapts clinical decisions based on observed outcomes as the disease progresses. An example for possible conflicting treatment options is: rapid recovery with higher dose for better short term outcome, manageable dose with slow recovery for longer survival. The reliance of the traditional rule-based and supervised clinical decision models on immediate outcomes, and their inability to learn the consequences of sequential decision making on long-term outcome, makes them unsuitable for personalized treatment planning. In contrast, Reinforcement Learning (RL), a subfield of ML when combined with multi-objective optimization, is well-suited to this sequential multi-objective task. It explicitly models decision-making in dynamic, sequential environments where optimal actions depend on evolving patient states aiming at long-term outcomes. This review examines the body of literature that applies RL and its variants to personalized treatment, and makes the following key contributions to the field:

- Background of different types of reinforcement learning and its key concepts is provided to support the readers who are new to this interdisciplinary field.

- Survey on reinforcement learning, deep reinforcement learning, and multi-objective deep reinforcement learning approaches for personalized treatment.
- Provides a theoretical and conceptual multi-modal multi-objective deep reinforcement learning framework that outlines the essential inputs, emphasizes the various objectives that need to be optimized, and highlights the critical role of clinical review in shaping the reward vector.
- Identification of several research challenges and potential future directions to advance the development of a viable personalized treatment model.

The successful implementation of this framework relies on the integration of multiple technologies across diverse domains and is discussed in Sect. 2, an abstract view of AI in personalized treatment is presented in Sect. 3. The survey methodology is defined in the Sect. 4. The survey mainly focuses on the application of RL to personalized treatment and the background of RL, Deep RL and MORL is discussed in Sect. 5. Section 6 presents the various research works based on personalized treatment strategies. A few of the commonly used and publicly available datasets are summarized in Sect. 7. Section 8 presents the key insights and synthesis drawm from the reviewed studies. Section 9 proposes a theoretical Multi-modal Multi-objective Deep RL framework for personalized treatment. The research challenges and future work is outlined in Sect. 10. The overall findings are concluded in Sect. 11.

## 2 Key technologies enabling personalized treatment

The progress in the field of genomics, proteomics, digitization of healthcare data, Internet of Things (IoT) and advancement in high computing facilities, Artificial Intelligence (AI) are a few of the key technologies that have enabled the development in the field of personalized treatment. The brief explanation of these technologies is provided below to support the readers who are new to this field. The role of AI is discussed in the subsequent section.

- *Genomics*: Genomics is the study of individual's entire DNA sequence and is useful in understanding genetic makeup. Variations in the DNA sequence give rise to genetic variations which are closely associated with individualized responses to medications. Pharmacogenomics uses the genomic information of a patient to predict drug dosage, prevent harmful drug reactions, promote drug discovery, evolution of targeted drugs for a disease, predicts the activation of pro-drug, etc. Thus, genomic data is the fundamental requirement which helps the clinicians to select an individualized drug and dosage based on their genetic information. Advances in genomics has accelerated the development of personalized treatment strategies that aim to develop highly precise and patient-specific treatment strategies [11].
- *Proteomics*: Proteomics is a study on protein sample to understand the structure and function of a protein, protein-protein interaction, and protein expression, etc. Proteomics is useful in identifying the protein biomarkers that enable early identification of disease, monitoring of disease progression, and the tailoring of treatment strategies. Protein biomarkers are crucial in targeted drug discovery, in which the drugs target the patient-specific protein associated with a given disease.
- *Digitization of electronic health records*: The personalized treatment needs the time-dependent patient-level data including the clinical observations, treatment

interventions, and outcomes for individual patients which is referred to as longitudinal data. The longitudinal patient data obtained from the electronic health records (EHRs) constitutes another mandatory data requirement of personalized treatment to enable the construction of patient trajectory-based model which is required for sequential-decision making. As AI models are data hungry, the availability of patient records in the digital form is one of the key enabler of personalized treatment, by enabling large-scale data analytics and data-driven clinical decision making.

- *Internet of Things (IoT)*: IoT facilitates real-time patient data collection using wearable medical device. The continuous data generated by these devices provide timely and up-to-date information on patient's status and outcomes thereby supporting personalized treatment planning.

To continue, the transition from "one-size-fits-all" to "patient-specific medicine" requires close collaboration between various stakeholders and the integration of various technologies, as depicted in Fig. 1. Stake holders across diverse sectors including government bodies, pharmaceutical and biotechnology companies, researchers from technical, clinical and genetic domains, AI engineers, insurance companies, etc., need to collaborate to develop a robust personalized therapy operating model. The success and findings of the human genome project have laid a strong foundation for advancing the idea of personalized or molecularly-targeted therapies. To further accelerate progress, both the government and industries should actively coordinate, support and encourage the researchers and healthcare sectors under the theme of personalized treatment through various schemes.

## 3 AI in personalized treatment: an abstract view

Personalized treatment can be viewed as a closed loop model with two broad stages and an abstract representation is depicted in Fig. 2. The pipeline starts with an acquiring of data from wearable IoT devices, clinical investigations, and collection of data from EHRs, etc. The first stage involves analysing the patterns from the high-dimensional comprehensive patient data to characterize the current state of the patient. The second




**Fig. 1** Stake holders of personalized treatment

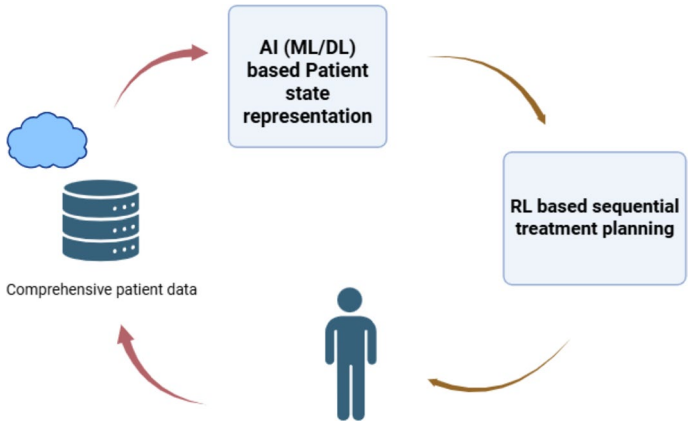



**Fig. 2** AI in personalized treatment

**Table 1** A few research works on state representation in RL for personalized treatment

| References | Data | Algorithms |
|---|---|---|
| [12] | MIMIC-III (time varying vital and lab measurements, demographic features) | Sequential representation using recurrent modeling approaches |
| [13] | MIMIC-III (features relevant to blood anti-coagulation) | Transitional variational autoencoders |
| [14] | MIMIC-III, dbSNP and ClinVar (genomic data), wearable IoT devices | Adaptive Temporal-Fusion |
| [15] | TCGA | Large-scale constrained variables-based multitasking co-operative evolutionary algorithm |

stage focuses on optimal treatment planning based on the representation derived in the first step. The following section provides an overview of suitable AI methodologies for implementing these two stages. Machine Learning (ML) and Deep Learning (DL), the sub-fields of AI are extensively applied at the academic as well as industry levels to address a wide range of healthcare problems like data analysis, image classification, early prediction of a disease, volumetry and biometric measurements, etc. In conventional practice, the supervised models are trained and validated for a specific task using a pre-existing, task-specific datasets prior to deployment. Following deployment, these models make static predictions based on previously learned parameters. To account for changing data distributions, concept drift, and changing clinical settings, such models are periodically retrained or finetuned from saved checkpoints on new batch of data. These models are more suited in representing the current state of the patient from the comprehensive patient data. The Table 1 summarizes a few of the works focusing on state-representation in the personalized treatment.

In personalized treatment planning, the objective of the second stage is to decide the suitable intervention plan, which is inherently a sequential decision making process, as each treatment intervention depends on the patient's current state, which evolves based on previous treatment response. To effectively support and enhance this sequential decision-making process, a sequential-learning algorithm capable of adapting to individual patient trajectories over time is required. As stated before, RL is particularly well-suited to this setting, as it models decision-making in dynamic, sequential environments where optimal actions depend on evolving patient states and aim to optimize long-term outcomes. A classical RL learns or gets trained through the trial-and-error method through

continuous interaction with the real-time environment, without relying on a labeled dataset, and is referred to as online-RL. This is best demonstrated by the success of RL in AlphaGo Zero by Google DeepMind [16] which was trained from scratch using self-play RL without any human demonstration data. But, online-RL needs a large number of interactions to converge, making it time-consuming and is infeasible in safety-risk applications like healthcare, autonomous driving, finance, etc. where trial-and-error exploration may be highly-risky, dangerous, unethical, and expensive [17, 18]. Consequently, RL based personalized treatment planning systems are off-line trained from historical patient trajectories with constrained exploration, or alternatively using a simulator that emulate a real-world environments for exploration. Thus, the RL models are initially trained offline, assessed using off-policy evaluation, and are then fine-tuned in on-line mode on actual environmental settings under the supervision of an expert to ensure safety and clinical validity. Together, AI-based accurate patient state representation in the first stage and sequential decision making in the second stage enable a dynamic, closed-loop model for personalized treatment.

## 4 Survey methodology

The study is a narrative review of RL based approaches in personalizing the treatment strategies. The review aims to synthesize the methodological developments with an emphasis on:

- Design patterns in state, action, and reward formulation
- Clinical domains in which RL/Deep RL (DRL)/Multiobjective Deep RL (MODRL) are applied
- Adoption of evaluation strategies and safety mechanisms
- Integration of multi-objective optimization techniques
- Various datasets employed in existing studies

Accordingly, literature search covering studies published between 2020 and 2025 was conducted across major academic databases and indexing platforms, including Google Scholar, Scopus, Web of Science, IEEE Xplore, Springer, and PubMed. Boolean search filters were applied with various combinations of the following keywords ("multi-objective" OR "reinforcement learning" OR "deep reinforcement learning" OR "offline reinforcement learning" OR "safe reinforcement learning" ) AND (“customized care" OR "patient-centered treatment" OR "personalized treatment" OR "precision medicine" OR "patient-centered" OR "Dynamic treatment regimes" OR " safety-risks" OR "optimized treatment" OR "individualized treatment) OR (("health" OR "healthcare" OR "clinical") AND ("treatment" OR "medicine" OR "diagnosis")). These studies were then subjected to a full-text review for eligibility, focusing on methodological rigor and relevance to personalized treatment approaches with more emphasis on the papers published in 2024 (15 articles) and 2025 (28 articles). Review articles, editorials, general healthcare based applications, non-RL based papers are excluded in the study. The detailed study selection process is summarized in the PRISMA-style flowchart as given in Fig. 3. Out of 280 records identified, only the records that met all the inclusion criteria detailed below are included in this narrative review.

1. English publications.
2. Full-text articles are available.

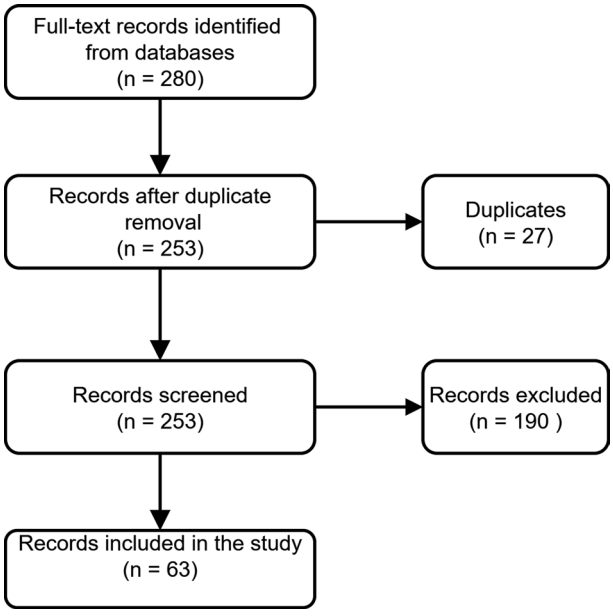



**Fig. 3** PRISMA-style flow diagram outlining the structured literature search and screening process adopted for this narrative survey

**Table 2** Comparison of existing review papers with the proposed survey

| Review paper | DRL and MODRL background | survey on MODRL and MOO methods | Framework | Safety/OPE | Integration of multi-modal data and multi-objective reward design |
|---|---|---|---|---|---|
| Banumathi et al. [19], 2025 | × | × | × | Limited | × |
| Timothy et al. [20], 2025 | × | × | × | Limited | × |
| **This work** | ✓ | ✓ | ✓ | ✓ | Detailed |

3. Latest studies reporting the use of RL related techniques in methodology.
4. Studies reporting on clinical decision support systems for personalized treatment.
5. Studies focusing the deployment related methodology and challenges in RL for healthcare

Table 2 summarizes the contribution of the present study in comparison with the existing survey papers (only full-text available articles considered) by Banumathi et al. [19] and Timothy et al. [20]. Unlike previous studies, the present study provides a comprehensive background on RL, DRL, MOO, and MODRL methods to support the readers who are new to this interdisciplinary domain.

To the best of our knowledge, this is the first survey on personalized treatment that emphasizes the necessity of optimizing multiple potentially conflicting clinical as well as non-clinical objectives using multi-modal, high-dimensional, longitudinal patient data.

## 5 Preliminaries

In simple terms, RL interacts in the environment, gets feedback in the form of rewards, and learns to make optimal decisions. DRL is the extension of RL that uses deep neural network to deal with complex problems. Multi-objective DRL is a variant of DRL integrated with Multi-objective optimization, and is used to deal with problems having

multiple conflicting objectives. This section presents the fundamental concepts of RL, DRL and MORL necessary to understand their application to personalized treatment.

### 5.1 Reinforcement learning

RL is based on the Markov Decision Process (MDP) [21] and is characterized by a set of states (S), actions (A) and corresponding rewards (R). The present inputs decide the current state $s_t$ of the learning agent, and the next action $a_{t+1}$ is decided by its past experience as well as the present state. The agent gets back positive or negative rewards $r_{t+1}$ based on the outcome of the action taken as depicted in Fig. 4. Based on the reward and external inputs, the next state $s_{t+1}$ is decided. Thus, the RL framework exactly matches with the healthcare operations and RL is been applied extensively to solve numerous healthcare problems [22].

Research in RL based solutions for real-world healthcare applications is still evolving, and the suitability and significant potential of RL underscore the need for continued investigation in this area. RL aims at enabling agents to learn optimal behavior through interaction with their environment, receiving feedback in the form of rewards with the goal of maximizing cumulative rewards over time. RL places a strong emphasis on learning by experience, in contrast to supervised learning, which depends on labeled datasets, or unsupervised learning, which reveals hidden patterns within data. Without the need for labeled supervision, an autonomous agent experiments with different activities and learns from the feedback it receives, which might take the form of positive or negative reward.

By determining which action sequences produce positive results in various scenarios, the agent iteratively improves its strategy. Over time, it learns to operate effectively with minimal or no human intervention. Rather than being guided by immediate rewards alone, the agent also takes into account delayed or long-term rewards when evaluating the effectiveness of its decisions. Ultimately, RL enables the agent to discover an optimal policy, a strategy to choose actions that yield the highest expected reward over time. The core components of RL include:

- *Agent*: The decision maker that interacts with the environment by taking actions. By optimizing the total reward it obtains from these encounters, the agent seeks to gradually enhance its behavior. In healthcare, the agent could be a clinical decision support system that recommends treatments.
- *Environment*: The external environment with which the agent engages. It delivers feedback to the agent through changes in state and associated rewards. For instance,

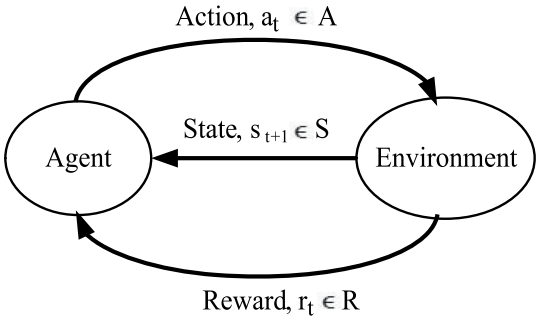



**Fig. 4** Agent–environment interaction in reinforcement learning

a patient's physiological reactions to treatments could be considered part of the environment in a medical setting.

- *States* (*S*): Represented as $s \in S$, it is a complete description of the environment at a particular time. In medical applications, a state might represent the clinical status of a patient. In a clinical context, a state might include patient vitals, lab results, medical history, and current symptoms.
- *Actions*(*A*): Represented as $a \in A$, it is a decision or intervention that the agent can take. Actions might be continuous (dosage quantities) or discrete (prescribe medication). Each action influences the environment and affects future states and rewards.
- *Policy* ($\pi$): Represented as $\pi(a \mid s)$, a policy describes the agent's behavior by mapping each state to a probability distribution over possible actions. It guides decision-making during the learning process by estimating the possibility that an agent will choose a specific action when in a particular state.
- *Reward*: A scalar feedback signal *R*(*s*, *a*) that indicates how good or bad an action is in a given state and denotes the immediate reward obtained upon executing an action $a$ in state $s$. This feedback guides the agent's learning by evaluating the desirability of the action's outcome. The agent aims to maximize the cumulative reward collected over time. In healthcare, a reward could be a positive signal when a treatment recommendation leads to improved patient health outcomes.
- *Value function* ($\mathrm{V}(s,a)$ *or* $\mathrm{Q}(s,a)$): Measures the expected cumulative reward that an agent can attain while following a specific policy, starting from a given state (or state-action pair). $\gamma \in [0,1)$ is the discount factor that determines the importance of future rewards.*Markov decision process*: A Markov Decision Process (MDP) [23] provides a mathematical framework for modeling decision-making in scenarios characterized by partial stochasticity and agent control. By interacting with the environment over time, this formulation allows the agent to learn optimal behaviors. Formally, a MDP is defined by the tuple $(S, A, P, R, \gamma)$. The objective of the agent is to learn a policy, that maximizes the expected cumulative discounted reward by mapping states to a distribution over actions as shown in Eq. 1:

$$G_t = \mathbb{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \right] \tag{1}$$

Eq. 1 represents the expected cumulative discounted reward received from time step $t$ onward, under a policy $\pi$. The components are:

- $G_t$: Total return starting at time step $t$.
- $\mathbb{E}_\pi[\cdot]$: Expectation under policy $\pi$, which governs action selection.
- $\sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$: Discounted sum of future rewards.
- $\gamma \in [0,1)$: Discount factor that weighs immediate rewards more heavily than future ones.*Value-based and policy-based Reinforcement learning*: Value-Based RL focuses on estimating value functions that represent the expected return of states or state-action pairs. The agent implicitly derives an optimal policy by selecting actions that maximize these value estimates. Common approaches include Q-learning and its deep variant, the Deep Q-Network (DQN) [24], where the optimal policy is obtained

through greedy action selection with respect to the learned action-value function $Q(s, a)$.

Policy-Based RL, in contrast, involves directly learning a parameterized policy, which maps states to action probabilities. It uses strategies like the REINFORCE algorithm oractor-critic methods to maximize the predicted cumulative reward and thereby improve the policy. Policy-based methods can learn stochastic policies that better manage exploration and are especially well-suited for high-dimensional or continuous action spaces.

*Model-based and model-free reinforcement learning*: Model-free RL algorithms learn optimal strategies through continuous trial and error approaches with the environment, without any prior information about the environment model. Some healthcare applications use model-free RL approaches such as DQN, SARSA [21], Soft Actor-Critic (SAC) [25], Proximal Policy Optimization (PPO) [26], etc. The model-based approaches learn the environment model and then use the model to predict the future outcomes before deciding the next action. Based on the outcome of the action taken, the policy is updated. Monte Carlo tree search, model predictive control, dynamic programming, etc., are a few well-adopted model-based approaches.

The choice of the RL algorithm for a given problem also depends on whether the state-action space is discrete or continuous, low-dimensional or high-dimensional, online or offline training, prior information of the environment is known or not. The basic RL approaches are most effective in environments with limited state-action space. This limitation has led to the development of DRL algorithms and has been widely used in applications such as autonomous driving [27], recommendation systems [28, 29], games [30], etc. The next section provides a brief discussion of DRL methods.

### 5.2 Deep reinforcement learning

The state-action space of most real-world problems is continuous and high-dimensional in nature. Therefore, a tabular form representation of $V(s, a)$ or $Q(s, a)$ is no more suitable and is replaced with deep neural networks, as shown in Fig. 5. This led to the development of DRL algorithms.

Botvinick et al. [31] highlight that by combining RL with deep neural networks, DRL makes it possible to handle high-dimensional inputs such as images and intricate sensory data, whereas traditional RL depends on predefined features and simpler function

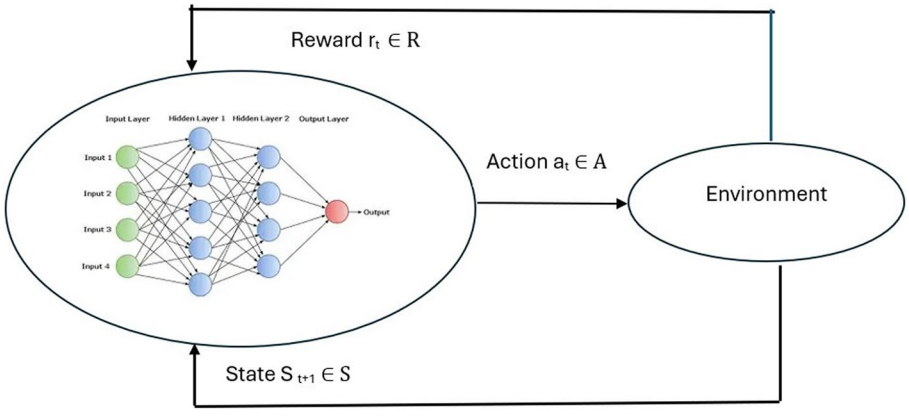



**Fig. 5** Deep reinforcement learning framework

approximations. End-to-end learning from raw inputs to actions is made possible by this integration and has been widely used in applications such as autonomous driving [27], recommendation systems [28, 29], games [30], etc.

DRL systems leverage deep neural networks, specifically convolutional and recurrent architectures, to autonomously learn hierarchical and abstract feature representations of high-dimensional input spaces (e.g., visual, temporal, or multimodal data). This hierarchical representation learning enables DRL agents to extract low-level patterns, (such as edges or movement) and gradually compose them into higher-level concepts (like object categories, spatial relationships, or temporal sequences), facilitating generalization across varying environments or tasks. In particular:

- *Convolutional neural networks (CNNs)*: [32] define CNNs as a specialized class of artificial neural networks that have demonstrated superior performance across a wide range of computer vision applications. They are designed to automatically and adaptively learn spatial hierarchies of features from input data through backpropagation, using a structured architecture that includes convolutional layers, pooling layers, and fully connected layers.
- *Recurrent neural networks (RNNs)*: In order to promote thinking across sequential decision-making processes, RNNs, which are neural networks built for sequential data with cyclic connections, capture temporal dependencies.
- *Transformer-based models*: By enabling agents to flexibly focus on task-relevant information across time and space, attention-driven architectures, which describe dependencies without recurrence, improve DRL and successfully capture long-range interactions. Transformers, which were first created for Natural Language Processing (NLP), are excellent at processing sequential data and are being used more and more in DRL to handle challenging decision-making tasks involving contextual and temporal relationships.

### 5.3 Multi-objective optimization and multi-objective deep reinforcement learning

Majority of the practical applications have *n* number of multiple conflicting objectives $O_1, O_2, ...O_n$ and are MO in nature. Two established and well-adopted strategies for MOO are: reformulating MOO to single-objective optimization type also referred to as scalarization, and the other method is pareto optimization or vector optimization. The commonly used types under the former method are: weighted sum method [33], or the constraint method [34], or max-min [35] method which is very popular in ML and Generative Adversarial Networks (GAN), sequential method, etc. Let $O_i(x)$ be the mathematical representation of the $i^{th}$ objective function which represents a specific metric to be optimized, $x \in X$ is a design variable selected from a set of $X$ on which the objective functions depend. Each objective function is assigned a scalar weight $w_i \leq 1$ satisfying $0 \leq w_i \leq 1$, with the constraint $\sum_{i=1}^{i=n} w_i = 1$. The weighted sum method uses a linear or convex combination of weighted objective functions as shown in Eq. 2.

$$Y = \sum_{i=1}^{i=n} w_i O_i(x) \tag{2}$$

The constraint method aims at maximizing one objective function $O_i(x)$ with a constraint $C_j$, $i \neq j$ on all remaining objectives, as shown in Eq. 3:

$$Maximize : O_i(x),\ such\ that\ O_j(x) \geq Cj, j = 1, 2, 3, ..n\ and\ j \neq i \tag{3}$$

The max-min method provides a solution that aims to maximize the performance of an under-performing objective as shown in Eq. 4:

$$Maximise\{min_{i=1,2,3,..n}\ O_i(x)\} \tag{4}$$

In the sequential method, the objective functions are optimized in a hierarchical manner, where the most important objective is optimized first. Subsequent objectives are then optimized one at a time, in decreasing order of priority, subject to the optimal values obtained for the higher-priority objectives. Let the priority order be $O_1 > O_2 > O_3 > .. > O_n$. In the first step, $O_1(x)$ is optimized as shown in Eq. 5

$$min_{x \in X}\ O_1(x) \tag{5}$$

In the second step, $O_2(x)$ is optimized, where $O_1(x) = O_1max$. The process is thus repeated in lexicographic order.

Pareto optimization finds a set of pareto-optimal solutions $s$ that cannot be maximized or improved further to $s'$ without compromising the performance of at least one objective, that is $O_i(x)_{s'} \geq O_i(x)_s$ and the plot or curve of all pareto-optimal solutions is pareto front as shown in Fig. 6

MOO uses statistical approaches to optimize multiple conflicting objectives. DRL, on the other hand is well-suited for handling high-dimensional data and learns optimal policies through continuous interaction with environment. The integration of these two methods, makes it possible to develop context-aware policies using a DL framework that effectively optimizes multiple conflicting objectives. Accordingly, MODRL uses a reward vector where each component in the vector corresponds to the performance of a policy with respect to a specific objective, as shown in Fig. 7.

## 6 Personalized treatment strategies

With advancement in the field of RL, researchers have applied various RL algorithms to find the optimal results in personalizing the therapy approaches. The review of the literature reveals that some studies concentrate on traditional RL approaches, as detailed in Subsection 6.1, while many researchers have specifically explored DRL algorithms, as examined in Subsection 6.2, and a very few have explored MODRL as outlined in Subsection 6.3.

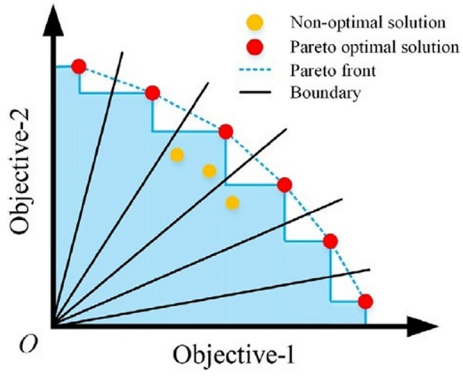



**Fig. 6** Pareto optimization [36]

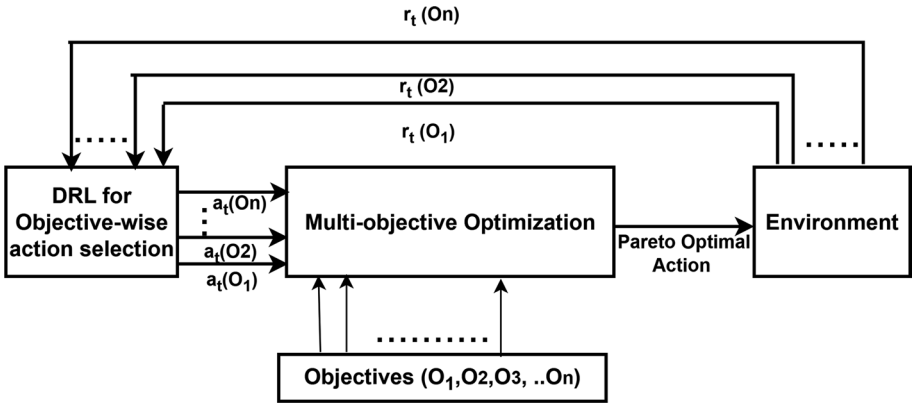



**Fig. 7** Multi-objective deep reinforcement learning with reward vector

### 6.1 RL in personalizing the treatment

Tsoukalas et al. [37] offer a probabilistic framework designed to improve clinical judgment when treating sepsis. To infer patient states, forecast potential trajectories, and suggest the best antibiotic combinations, the researchers utilized the EHR of 1492 adult Intensive Care Unit (ICU) patients who satisfied at least two criteria of Systemic Inflammation Response Syndrome (SIRS) and used a Partially Observable Markov Decision Process (POMDP). With an emphasis on antibiotic administration protocols, they developed states, behaviors, observations, and incentives based on expert knowledge and clinical practice. The study showed that adherence to the model-based antibiotic policy resulted in a higher percentage of favorable patient outcomes, compared to when alternative therapies were employed. Killian et al. [12] conducted the inaugural rigorous empirical evaluation of sequential state representations for RL in healthcare, utilizing the Medical Information Mart for Intensive Care (MIMIC-III) [38] sepsis dataset comprising 19,418 patients, they compared seven encoding architectures, including RNN autoencoders, Neural Controlled Differential Equations (CDE), and deep signature transforms, to predict subsequent observations and devise treatment policies. Authors used off-policy batch constrained Q learning to learn optimal policies from patient state represented using sequential networks and is evaluated using weighted importance sampling. Notably, representations based on Neural CDE significantly surpassed alternative approaches in policy learning using discretized Batch Constrained Q-learning. This study demonstrated that sequentially formed state representations effectively facilitate policy learning in batch settings, underscoring the importance of respecting the sequential and partial nature of healthcare data.

The effectiveness of RL algorithms in healthcare is fundamentally contingent upon the quality of state representations derived from complex, noisy, and inherently partial clinical observations. Wang et al. [39] demonstrated that RL can effectively generate high-quality synthetic medical images by leveraging radiologists' evaluations as a reward signal for policy learning. The proposed model has shown promising improvement in the identification of targeted therapy-sensitive EGFR mutations. Quan et al. [40] have modeled feature selection as a probabilistic learning process and have proposed statistical RL based explainable feature selection framework enabling transparent decision framework.

Yoon et al. [41] employed a tree-based model to identify optimal personalized treatment strategies for hand deformities in patients with severe rheumatoid arthritis. Analyzing data from 169 adults with significant ulnar deviation and extensor delay, they found that silicone metacarpophalangeal arthroplasty was beneficial for patients over 55 years of age, those with low grip strength ($< 12kg$) and those with severe deformities. The study suggests that T-RL can guide surgical decision-making by integrating patient-specific factors to improve outcomes such as pain, function, and quality of life. Baek et al. [42] propose DecompPI, a novel model-free policy optimization algorithm for personalized digital health interventions and a case study was conducted on tuberculosis treatment adherence. Ayyalasomayajula et al. [43] explore personalized treatment of diabetic foot ulcers using advanced ML and RL methods. By combining ensemble models, deep learning for image and sequential data, and Q-learning for adaptive treatment policies, the study demonstrates improved diagnosis, prediction, and patient outcomes with reduced complications and costs. Zhou et al. [15] explore the use of the RL algorithm to personalize Cognitive Behavioral Therapy (CBT) for anxiety treatments. The authors used high-dimensional synthetic data to model the psychological state of the patient with various CBT techniques and interventions that represent possible actions. By employing a continuous action and reward framework, the RL agent learns to optimize treatment strategies, leading to improved therapeutic outcomes.

Gayathri et al. [14] introduce the Personalized AI-Driven Therapeutic Framework (PAI-DTF) that combines multi-modal data fusion (genomics, EHRs, wearable sensors), a Hierarchical Deep Neural Network with Dynamic Biomarker Importance module for adaptive feature selection, and Patient-In-the-Loop Reinforcement Learning (PIL-RL) that incorporates real-time patient feedback and treatment adherence to dynamically optimize therapy over time. The PIL-RL component allows the system to continuously adapt treatment plans based on patient responses, biomarker changes, and adherence patterns, moving beyond static protocols to patient-centric adaptive care. The framework achieves high accuracy and consistency in treatment predictions through hybrid and ensemble models, demonstrating strong potential for real-world clinical applications in personalized precision medicine.

### 6.2 DRL for personalized treatment

Deep reinforcement learning extends traditional RL by using deep neural networks to handle high-dimensional clinical data such as complex EHR records, continuous physiological signals, and medical images, with methods ranging from Value-Based DQN for discrete treatment decisions to policy-based methods like PPO for continuous control tasks like adaptive deep brain stimulation and Off-Policy Actor-Critic approaches like Deep Deterministic Policy Gradient (DDPG), for personalized vital signs management.

Researchers have explored and proposed different DRL methods to recommend a personalized treatment model. The various DRL methods used by the authors are listed in Fig. 8.

#### *6.2.1 Policy - based DRL*

Mulani et al. [44] discusses the use of DRL to offer personalized healthcare advice, such as suggestions for medications, consultations, nutrition, and exercise. It presents an Off-Policy Actor-Critic model that adjusts its strategies in real-time based on user feedback,

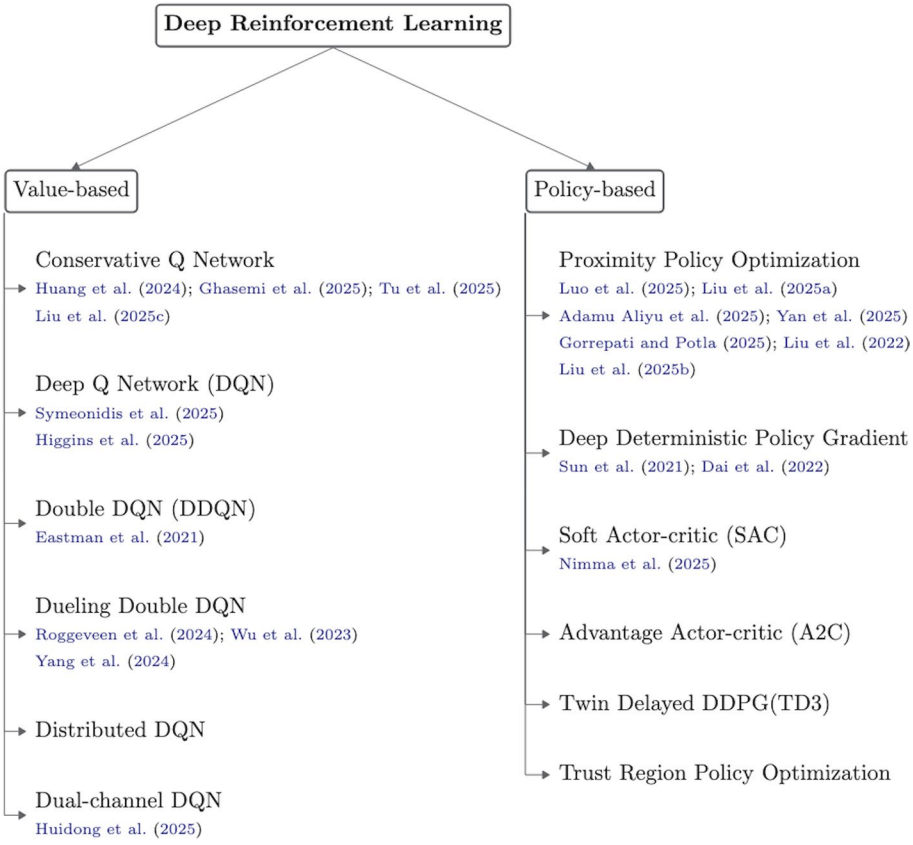



**Fig. 8** Taxonomy of included studies based on Deep Reinforcement Learning algorithms

effectively responding to changing user preferences and health situations. By framing the recommendation process as a series of decisions, this method captures the evolving needs and feelings of users over time. The chapter highlights the importance of combining data from wearable devices, EHRs, and nutrition information, analyzed through big data techniques, to create tailored recommendations.

Sun et al. [45] present a framework that uses RL to personalize the vital sign targets, such as blood pressure and oxygen saturation, in ICU patients. It employs a continuous action space RL algorithm, enhanced through supervised learning, to improve decision-making in critical care. The study utilizes the MIMIC-III dataset, a publicly available ICU database containing time-series physiological data, as the basis for training. Instead of using a traditional simulator, the authors built a data-driven patient simulator by modeling patient responses based on historical data from MIMIC-III.

Baucum et al. [13] present a generative neural network architecture called Transitional Variational Autoencoders (tVAEs), to create synthetic patient trajectories by learning direct mappings between patient states. Authors used A3C algorithm to learn the heparin dosage policy for ICU heparin patients. The agent was trained online using the environment modeled by tVAEs. The study shows that adding tVAEs significantly enhances the neural network's capacity to forecast the course of a genuine patient's illness.

Dai et al. [46] introduce an innovative Off-Policy framework that combines DNNs with RL to model and enhance healthcare interventions. The authors created a virtual human body simulator utilizing deep neural networks, which produces high-dimensional picture representations of latent health states in reaction to diverse interventions. To train

this simulator, they utilized a dataset comprising 990 tongue images categorized into nine body constitution types. Using a conceptual alignment DDPG method, the treatment strategy module helps the system learn efficient intervention policies guiding the simulated body from an unhealthy to a healthy condition. This closed-loop system shows how integrating DRL and simulation methods can help guide individualized healthcare plans. Huidong et al. [47] have proposed a Dual-channel Batch Constrained Deep-Q Learning (DBCDQ) to decide the personalized drug dosage in 0 to 5 scale under vasopressure and intravenous fluid treatment for ICU admitted sepsis patients. Patient state is represented as a function of features from static or demographic data, lab reports, and physiological data. Reward is designed as a function of the Sequential Organ Failure Assessment (SOFA) score and lactate levels. The reward value increases with a decrease in SOFA score and lactate levels indicating a sign of patient recovery. Authors have validated the proposed model using MIMIC III dataset.

While off-policy actor-critic (DDPG, SAC) achieves higher sample efficiency through experience replay, on-policy actor-critic (PPO, A3C) offers greater training stability by learning from its own experiences but requires more samples. Liu et al. [48] propose PPORank (Proximal Policy Optimization Ranking), an on-policy actor-critic framework that models personalized cancer drug recommendation as a Markov decision process, ranking anticancer drugs sequentially based on their predicted effects for individual cell lines using patient-specific molecular and clinical features. The PPO-based model, formulated with hidden state representations and evaluation signals to reduce variance, learns to continuously adapt drug selection over time as data accumulate, trained on two large-scale cancer cell line datasets (GDSC and CCLE). Experimental results demonstrate that PPORank significantly outperforms state-of-the-art supervised learning methods in precision drug ranking, showing that deep RL approaches have substantial potential for personalized treatment recommendations in precision oncology. Through an ablation study, Luo et al. [49] identified PPO to perform better than DDPG and actor-critic networks in administering the drug concentration for cancer patients. Authors modeled the interaction among the normal, tumor and immune cells using commonly used ordinary differential equations. Authors introduced noise during the training in order to strengthen models adaptability and robustness against real-time uncertain environments.

Yan et al. [50] optimize the dosing patterns of radiotherapy for head and neck cancer patients so as to reduce the toxicity or long-term side-effects. Tumor is modeled using linear quadratic model that relates radiation dose to the tumor cells. The current fraction of dose, the tumor volume, difference between personalized dose and conventional dose represent the present state of the patient. based on the current state, PPO agent decides the next action that is the fraction of the dose for the $\mathrm{i}^{th}$ day. The outcome of the treatment is measured using the amount of reduction in the tumor volume due to the personalized dose. Liu et al. [51] have combined the ResNet50 based classification model with PPO-based DRL model topredict transcription factor SRY box 9 (SOX9) in hepatocellular carcinoma patients. The features extracted from the classification model is given as state input to the generator model of PPO-clip. In subsequent iterations RL model generates weight distribution that highlight the portion of the CT image with SOX9 expression. The classification accuracy value of greater than 0.9 gives a positive

reward, and less than 0.9 gives a negative reward. Authors have used PPO-clip algorithm to ensure the policy updates within the range.

Zhao et al. [52] developed an adaptive Deep Brain Stimulation (DBS) computational model using PPO, to modulate stimulation currents in Izhikevich neuronal models targeting the Globus Pallidus internus (GPi) for tremor suppression in Parkinson's disease. The PPO agent learned to desynchronize pathologically synchronous neurons by optimizing stimulation frequency and amplitude in real-time simulation, achieving effective tremor attenuation while analyzing the trade-off between model efficiency and energy consumption. Compared to standard fixed-frequency DBS used in clinical practice, the RL-based adaptive approach demonstrated superior performance by dynamically adjusting stimulation based on neuronal activity patterns. Gorrepati et al. [53] introduce a disease-agnostic RL framework that models treatment as a POMDP by utilizing multimodal clinical data, including EHR, laboratory results, and imaging. The framework is trained using PPO, an actor-critic approach with LSTM, and incorporates Variational Autoencoders (VAEs) for state encoding, action masking, and a model-based simulator for off-policy rollouts. Their sources consist of over 50,000 patient records, with references to the MIMIC-III database. Experiments conducted in simulated environments and with large patient cohorts yielded a remarkable 93.6% accuracy, an F1 score of 0.937, a ROC-AUC of 0.964, and a sensitivity of 94.2%. Additionally, there was a 12.6% reduction in treatment cycles observed, alongside strong policy stability and scalability to 200,000 synthetic records. This framework demonstrates enhanced safety compliance and treatment efficiency compared to supervised baselines and traditional RL approaches. Aliyu et al. [54] used PPO for predicting asthma attacks based on the patient's current state. The current state includes the environmental factors, demographic data, the asthma test score, frequency of outdoor activities. Agent gets back a reward of +1 for correct prediction and -1 for wrong prediction.

#### *6.2.2 Value - based DRL*

Eastman et al. [55] developed a novel RL approach for chemotherapy scheduling that exhibited greater robustness to variations in patient-specific parameters than traditional optimal control methods. Utilizing a two-compartment tumor growth inhibition model for breast and ovarian cancers, they employed Deep Double Q-Learning to generate chemotherapy dosing schedules aimed at optimizing treatment efficacy while preserving bone marrow health. They trained their model using only nominal parameter values and tested it on 600 virtual patients with parameter perturbations of 15%, 20%, and 25%. A key innovation of their approach was the use of easily measurable relative bone marrow density as a feedback mechanism. This enabled the RL agent to personalize treatment schedules without the need for precise knowledge of patient-specific tumor growth parameters, which are often difficult to measure in a clinical setting. The results indicated that the RL-derived schedules maintained an optimality range of 92.5%–100% across all levels of perturbation, showing significant improvement over traditional optimal control methods, whose performance declined from 94.1% to 87.9% as parameter uncertainty increased. This study provides validation that RL methods can generate robust, personalized treatment protocols using only population-level parameter distributions and accessible biomarkers, offering a promising framework for clinical application when patient-specific model parameters are unknown.

Wu et al. [56] introduce the Weighted Dueling Double Deep Q-Network (DDDQN) with Embedded human Expertise (WD3QNE), a DRL model designed to enhance sepsis treatment strategies. The WD3QNE model integrates a novel target Q-value function with adaptive dynamic weighting to improve estimation accuracy and incorporates clinician expertise to guide decision-making, ensuring safer and more effective treatment recommendations. Additionally, the model employs a random forest algorithm for feature selection, enhancing interpretability by focusing on the most relevant clinical variables. Evaluated against state-of-the-art DRL methods using the MIMIC-III dataset, the WD3QNE achieved a survival rate of 97.81%, outperforming other models and demonstrating its potential to provide reliable, expert-informed treatment decisions for sepsis patients. To mitigate the issue of over-estimation of the Q values, Huang et al. [57] experimented with Conservative Q-Network (CQN) in which a regularizer is added during the training to reduce the overestimation and the trained algorithm performance is found to match more closely to clinical decisions than the DQN method. Authors applied CQN for optimizing the morphine dosage for the ICU patients. From the MIMIC III dataset, only the records of patients who are administered with at least one dosage of morphine and have the details on the intensity of patients pain (scale: 1–10) are used by the authors. The action is the range of morphine sulphate dosage in mg if morphine push treatment is chosen or morphine drip treatment is chosen. Kaur et al. [58] propose DQN with CNN model for optimizing the personalized drug dosage and timings. In addressing the persistent challenge of class imbalance in clinical decision-making, Yang et al. [59] proposed a RL–based framework, Q-imb, designed to improve prediction fairness and accuracy across multiple healthcare classification tasks. Q-imb, a DRL framework tailored for multi-class imbalanced healthcare data, transforms classification into a sequential decision-making task via a Markov decision process setup. They leverage dueling and double deep Q-learning architectures, augmented with a custom reward function and episode-based training, to prioritize correct classification of minority classes. The framework was evaluated on real-world clinical datasets including COVID-19 ER screening (binary task) and ICU diagnosis classification (multi-class task from the eICU-CRD) demonstrating more balanced and fair predictions, with notable improvement in minority-class detection over state-of-the-art imbalanced learning techniques. Q-imb consistently outperformed existing methods, confirming its effectiveness in mitigating class imbalance in high-stakes healthcare scenarios.

Roggeveen et al. [60] focus on optimizing ventilator settings for seriously ill COVID-19 patients using advanced RL techniques. Utilizing data from the Dutch ICU Data Warehouse, containing over 3000 patient records from 25 ICUs, the researchers trained 69,120 models based on the DDDQN architecture. The models' state space incorporated comprehensive clinical data such as ventilator settings, medications, and lab results, while the action space focused on adjustments to positive end-expiratory pressure and fraction of inspired oxygen. Furthermore, the use of policy constraints and the delta-Q metric offered valuable insights into how closely the model's recommendations aligned with clinicians' decisions, thereby improving the transparency and safety of applying RL in intensive care environments.

Tu et al. [61] introduce an offline, safety-focused RL model for optimizing sepsis treatment, addressing two key challenges: variable-length ICU episodes and sparse final rewards. The authors pre-process patient records from the MIMIC-III database to retain

variable-length time series and apply Conservative Q-Learning (CQL) to under-estimate rarely observed actions, thereby enhancing treatment safety. Crucially, they incorporate Apache II scoring as intermediate rewards to mitigate sparse terminal mortality signals, guiding decision-making more effectively. In experiments comparing CQL and DDQN with and without intermediate rewards, the CQL model augmented with Apache II achieved the highest evaluation score while maintaining action distributions closely aligned with physician behavior, and displayed superior robustness on out-of-distribution data. Nimma et al. [62] and Symeonidis et al. [63] predict insulin dosage levels for controlling the glucose level of the diabetic patients using a customized DQN model trained and tested on a MIMIC - III dataset. Higgins et al. [64] also used DQN model to recommend the personalized treatments for the patients with Helicobacter pylori infection.

The traditional off-policy RL method trained on the existing dataset tend to over-estimate the Q values as it learns the policy based on the previous experiences stored in the replay buffer. The over-estimation of Q-values may lead to sub-optimal strategies like overdose of drugs which may lead to toxicity in the patients. Liu et al. [65] present a DRL-based framework for optimizing outpatient rehabilitation in knee osteoarthritis, using a dedicated knee osteoarthritis bank (KOADB) and Random Forest feature selection to reduce the original 122 questionnaire items to the most influential predictors of treatment outcomes, thereby minimizing patient burden. Among the tested algorithms, Batch-Constrained Q-learning(BCQ) performed best (79.1%), surpassing real-world outcomes and demonstrating the potential of RL for personalized, adaptive treatment strategies in KOA management.

Ghasemi et al. [66] investigate obstructive coronary artery disease utilizing a large real-world registry (APPROACH; 41,328 patients) to learn offline RL policies specifically Q-learning, DQN, and Conservative Q-learning, structured as a MDP with k-means state clustering. They train on historical angiography encounters and assess the performance of RL versus physician decisions through off-policy evaluation using weighted importance sampling. The RL policies yielded an improvement in expected rewards of up to approximately 32% compared to physician policies regarding composite major adverse cardiovascular events (MACE), all while maintaining interpretable and clinically relevant policies.

Recent comprehensive surveys have examined deep reinforcement learning techniques in healthcare, comparing DRL methods.

Yu et al. [67] examines DRL for managing high-dimensional medical data, model-based techniques that employ environment simulations, and model-free methods like Q-learning and SARSA. Additionally, it discusses POMDPs for handling insufficient patient data and Inverse RL (IRL) for inferring expert decision-making. The survey illustrates that the optimal approach varies depending on the particular healthcare application; DRL performs better in complicated, high-dimensional scenarios, whereas model-based RL is helpful in organised environments.

Zhou et al. [68] examine how DRL techniques can be applied to various medical imaging applications. The authors categorise DRL applications into three main groups: optimisation tasks, which includes hyperparameter tuning, selecting augmentation strategies, and neural architecture search; parametric medical image analysis tasks, including landmark detection, object/lesion detection, image registration, and view plane

localisation; and other applications, such as surgical gesture segmentation, customised mobile health interventions, and computational model customisation. The authors stress that DRL can reduce computational loads by streamlining complex image analysis processes. This method works well for scaling to support high image resolutions without using a lot of memory. DRL also aids in striking a healthy balance between analytical precision and processing speed.

Khezeli et al. [69] examine how RL is utilized in healthcare, particularly action-value methods (Q-learning) and policy gradient methods, with applications in cancer, epilepsy, sepsis, and nephrology. The authors primarily discuss offline RL approaches learning from historical EHR and clinical trial data, emphasizing that action-value methods are suitable for discrete treatment decisions while policy gradient methods handle continuous dosing problems. Job et al. [1] explores DRL methods to optimize treatment strategies for critical patients, particularly focusing on dopamine and vasopressor administration. It emphasizes on state representation as a critical factor in enhancing the RL model's ability to derive optimal treatment strategies, and explores different configurations, particularly LSTM and LSTM-GRU, to encode patient states effectively. It compares different DRL architectures, including DDQN, DDDQN, showing that DDDQN achieves superior policy segregation and reward accumulation. Experiments conducted on ICU and MIMIC-III datasets demonstrate that the proposed model outperforms baseline methods in deriving effective treatment policies, favouring moderate vasopressor/dopamine administration for better patient recovery. The results indicate that the proposed model performs well with synthetic data, achieving rewards similar to those obtained with real data, demonstrating its ability to generalize effectively.

A systematic overview of scholarly research on DRL applied to personalized treatment strategies is summarized in Table 3. From the review on various DRL methods, it is evident that majority of the studies used different variants of DQN and PPO. In contrast, different forms of DDPG and other actor-critic architectures remain underexplored. Moreover, DRL is particularly advantageous for addressing large-scale problems; however, the reviewed works have primarily focused on leveraging its capability to handle high-dimensional state–action spaces.

### 6.3 Multi-objective approaches for personalized treatment strategy

Modeling the personalized treatment problem as a multi-objective problem is essential rather than optional. The first Sect. 6.3.1 introduces readers to commonly employed methods for solving MOO problems and highlights the multiple objectives considered in the existing studies. The subsequent subsection discusses how these methods are integrated with DRL including aspects like design of the reward vector and evaluation of performance. However, only a limited body of research has explored personalized treatment recommendation in the framework of MODRL.

#### *6.3.1 MOO for personalized strategies*

Abdulrashid et al. [70] have investigated an evolutionary heuristic search method to identify the optimal dose of drugs and the scheduling strategy to minimize tumor cells ensuring host cell viability. The authors formulated the MOO problem by modeling chemotherapy treatment through a dynamic model that captures the interaction between host cells, administered drug, and tumor cells. The population dynamics of tumor and

**Table 3** Detailed summary of personalized treatment studies

| Refs | Method used | Dataset / simulator | Disease | Clinical action type | OPE | Safety |
|---|---|---|---|---|---|---|
| [37],2015 | Probabilistic decision model | Sepsis dataset | Sepsis | Optimal policy to administer antibiotic drug combinations | × | × |
| [41], 2024 | Tree-based RL | Clinical dataset | Rheumatoid arthritis | Treatment strategy selection: surgical or non-surgical | × | × |
| [42],2025 | Policy Optimization | Behavioral dataset | Behavioral health | Personalized intervention recommendation | – | – |
| [43], 2025 | ML/RL (Q learning) hybrid | Clinical dataset | Diabetic foot ulcer | Adaptive Treatment planning | × | × |
| [44],2020 | Actor-critic network | Health dataset | General health | Personalized health recommendation | × | × |
| [45],2021 | DDPG | MIMIC - III | Sepsis and heart disease | Vital parameter trajectory control to improve survival rate | × | ✓ |
| [46],2022 | DDPG | Simulated system | General healthcare | Closed-loop treatment intervention strategy | × | × |
| [47],2025 | Dual channel BCQ | MIMIC- III | Sepsis | Drug dosing recommendation from 25 combinations of Vasopressor and intravenous fluid | ✓ | × |
| [48],2022 | PPO | CCLE, GDSE - Training, TCGA - Validation | Cancer | Ranking based drug recommendation | × | × |
| [49],2025 | PPO | Clinical dataset | Cancer | Chemotherapy drug dosage administration | × | × |
| [50], 2025 | PPO | Simulation based | Head & neck cancer | Radiotherapy fractionation scheduling | × | × |
| [51], 2025 | PPO | CT dataset | Hepatocellular carcinoma | Prediction of SOX9 expression status in CT images | × | × |
| [52], 2025 | Deep brain simulation with PPO | Computational model | Parkinson's disease | Deep brain stimulation control by modulating the current pulses | × | – |
| [53], 2025 | PPO | Multi-modal synthetic dataset | General planning | Real-time treatment planning | × | × |
| [54], 2025 | PPO | Asthma dataset from weather asthma app | Asthma | Exacerbation management strategy | × | × |
| [55], 2021 | DDQN | Tumor simulator | cancer | Chemotherapeutic drug scheduling | × | × |
| [56], 2023 | Dueling DDQN | MIMIC - III | Sepsis | Drug dosing recommendation | ✓ | × |
| [57],2024 | CQL | MIMIC -III | Pain management | Pain management through drug (morphine push, morphine drip) dosage recommendation | ✓ | × |
| [60],2024 | Cross-OPE of DDDQN models | ICU dataset | Critical care | Ventilator setting optimization | ✓ | × |
| [61],2025 | CQL | MIMIC III | Sepsis | Safe drug dosing recommendation | ✓ | × |
| [62],2025 | Actor critic network | Healthcare dataset | General healthcare | Risk-aware treatment strategy | ✓ | ✓ |
| [63],2025 | DQN | MIMIC - III | ICU patients | Glucose control through insulin dosing | × | × |

**Table 3** (continued)

| Refs | Method used | Dataset / simulator | Disease | Clinical action type | OPE | Safety |
|---|---|---|---|---|---|---|
| [64],2025 | DQN | H. pylori dataset from European registry | Helicobacter pylori | Antibiotic regimen selection | × | × |
| [65],2025 | BCQN | Demographic and Knee related clinical data | Knee osteoarthritis | Rehabilitation treatment optimization | × | × |
| [66],2025 | CQL | CAD dataset | Coronary artery disease | Treatment decision optimization | ✓ | × |

**Table 4** MOO for personalized treatment strategy

| Article year | MOO algorithm | Objectives |
|---|---|---|
| [71] | Evolutionary MO gray wolf optimization | Minimize the cancerous cells, Minimize of drug concentration |
| [72] | Constrained MO evolutionary optimization with ranking-based fitness function | Minimize driver nodes, Maximize the multiple sets of drug-targets, Optimal solution is then chosen using ranking based fitness function |
| [70] | Evolutionary heuristic search methods | Maximize or preserve the host cells populations, Minimize tumor cells |
| [74] | Constrained multimodal MO evolutionary optimization | Minimize driver nodes, Maximize the multiple sets of drug-targets, Identifies equivalent solutions that differ in their protein configuration |

host cells is analyzed using a mathematical model that incorporates patient-specific biological parameters.

Chen et al. [71] used evolutionary MO gray wolf optimization with conflicting objectives of minimizing cancerous cells and minimizing drug concentration to maximize healthy cells and keep them above a desired threshold during treatment, as they help minimize cancerous cells. The authors propose an optimal treatment strategy with a minimum dose of drug and a minimum effect on healthy cells. Personalized drug target is the identification of the patient-specific target molecule from the patient's molecular profile on which the drug should act upon. Conventional structural Network Control Principles (NCPs) aim at controlling the personalized gene interaction network through a minimum number of driver-nodes. Liang et al. [72] used MONCP to provide multiple drug target sets and an increased solution space with detailed target information. The authors then developed a constrained evolutionary algorithm based on MOO with a ranking fitness function, to predict an efficient drug target personalized to the patient's genomic profile. The proposed algorithm was validated using a cancer genome atlas database [73]. Liang et al. [74] have extended the work to consider multimodal solutions. In this context, the authors have defined the existence of multiple equivalent solutions that have different points in the decision space and represent the same point in the solution space as multimodal. Through the multimodal approach, equivalent solutions that differ in their protein configurations are also identified. Finally, the overall summary of the existing MOO approaches is presented in Table 4.

#### 6.3.2 Multi-objective deep reinforcement learning for personalized treatment strategy

Due to the high interpretability and capability to generate multiple outcomes of decision trees, Song et al. [75] have proposed a multi-objective tree-based RL to provide multiple viable DTR solutions at every stage of treatment. The authors have validated the model using patient data from patients with prostate cancer undergoing chemotherapy. Huo et al. [76] have used the number of tumor cells, effected-immune cells and drug concentration to represent the present state of the patient, amount of drug dosage in the range [0,1] as the action, and difference in the number of tumor cells and immune cells as the reward. Authors have used a mathematical model proposed by Depillis et al. [77] to represent the tumor. Li et al. [78] have demonstrated the improvements treatment plan, prediction accuracy and reduction in time required for decision-making when MODRL with explainable AI is used instead of conventional methods. The state space represents patients state along with present treatment status, action space consists of possible clinical interventions in orthodontic-orthognathic treatment. Multi-objective rewards with reward for each specific clinical goals are combined into a single reward using a weight vector. The model is trained and validated using 347 retrospectively collected data. DeCarlo et al. [79] aimed at achieving precise dose of givinostat treatment for polycythemia vera patients using platelet count, white blood cell count, and hematocrit as three biomarkers. Accordingly, the authors considered normalizing these three biomarker levels as three objectives while modeling reinforcement learning. Research in this direction is still limited, indicating scope for further exploration.

## 7 Datasets

The implementation of RL based personalized treatment strategy requires the integration of multi-omics (genomics, proteomics, etc) data along with longitudinal patient data acquired from EHRs. The large-scale oncology initiative like The Cancer Genome Atlas (TCGA) [73] provides molecular as well as clinical profiles derived from cohorts of cancer patients that enable the study of disease characteristics and progression, and treatment outcomes at the population-level. In addition, Catalogue of Somatic Mutations In Cancer (COSMIC) [80] represents a curated repository of somatic mutations aggregated from multiple studies, whereas Cancer cell Line Encyclopedia (CCLE) [81] is a laboratory-generated cell-line dataset that provides detailed molecular and drug response information under controlled experimental conditions. These supporting datasets (TCGA, COSMIC, and CCLE) contribute to personalized treatment through treatment candidate selection, biomarker identification, and feature representation, thereby supporting accurate state representation and personalized action selection in RL framework. In contrast, the longitudinal datasets(ex: MIMIC III/IV) are essential for sequential decision making as they enable the modeling of treatment trajectories required for RL. At present, only a limited number of studies have integrated both multi-omics data and longitudinal patient-level data. The datasets mentioned above are publicly available research resources; however, access to certain datasets, such as TCGA and MIMIC, requires appropriate authorization or completion of data access procedures due to the presence of sensitive patient information. A brief overview of these datasets is provided below.

*Cancer cell line encyclopedia (CCLE)*:

Cancer cell lines are cancerous cells derived from tumors and cultured under controlled laboratory conditions for use in various areas of medical research. They are primarily employed to study cellular proliferation rates and to evaluate the efficacy of potential therapeutic drugs. Cancer cell Line Encyclopedia (CCLE) [81] is a dataset of large scale molecular profiles. The CCLE dataset includes around 1457 cancer cell lines with all their details which is essential for cancer related research and also the mutation details. This dataset is collected by Broad Institute in collaboration with Novartis Institutes for Biomedical Research. In the subsequent phases of the project, MD Anderson and Harvard Medical School joined as collaborators and CCLE data is currently available through the Cancer Dependency Map (DepMap) portal https://depmap.org/portal/. For each cell line, the dataset provides: (i) metadata (e.g., standard name, tissue source, cancer type, and subtype), (ii) gene expression profiles derived from RNA sequencing, (iii) somatic mutation data obtained from DNA sequencing, (iv) proteomic measurements, (v) drug sensitivity data, (vi) DNA methylation profiles, and (vii) copy number alterations, along with comprehensive multi-omics and phenotypic information.

*Genomics of drug sensitivity in cancer (GDSC)*:

Genomics of Drug Sensitivity in Cancer (GDSC) [82] is a publicly available large pharmacogenomic dataset useful in developing various predictive models. The dataset is also useful to identify a biological molecule that plays a significant role in the progression of cancer that can be targeted by the anti-cancer drug.

*The cancer genome atlas (TCGA)*:

TCGA [73] project provides comprehensive data on different types of cancerous tumor tissues collected from 11,000 patients and the processed data is available at Genomics Data Commons Portal https://portal.gdc.cancer.gov/. The project started in 2006 and is funded by the U. S Department of Health and Human Services. The dataset has information about gene-level expression, genomic variations, protein expression and annotations, etc.

*Medical information mart for intensive care (MIMIC)*:

Medical Information Mart for Intensive Care (MIMIC III and MIMIC IV) [38] are in use by the researchers. MIMIC IV 2.2 data was released in January 2023 and is the publicly available data of ICU or hospitalized patients collected in between 2008 and 2019. The data can be accessed from Physionet website https://physionet.org/content/mimiciv/2.2/ after the completion of ethical training. The hospitalized patients data module consists of lab laboratory measurements, microbiology cultures, provider orders, medication administration, medication prescription, hospital billing information, online medical record data, and service-related information. ICU-admitted patient module details include intravenous and fluid inputs, ingredients for the aforementioned inputs, patient outputs, procedures, information documented as a date or time, and other charted information.

*The catalogue of somatic mutations in cancer (COSMIC)*:

COSMIC [80] is a publicly available, comprehensive and manually curated database that catalogs somatic mutations associated with human cancers. It provides detailed information on a wide range of genetic alterations, including coding and non-coding mutations, gene fusions, copy number variations, and drug-resistance mutations. The database integrates data from multiple experimental studies and clinical reports, ensuring high-quality and well-annotated mutation information for cancer genomics research.

COSMIC is widely used for identifying actionable mutations, understanding cancer progression mechanisms, and supporting biomarker discovery in precision oncology. In addition, resources such as the Cancer Gene Census provide curated lists of genes implicated in cancer development. While COSMIC offers valuable insights into mutation-driven disease mechanisms and therapeutic relevance, it does not provide patient-level longitudinal treatment trajectories and is therefore primarily used for feature extraction and biological interpretation rather than direct reinforcement learning-based decision-making.

## 8 Key insights and synthesis

The temporal review of the literature has shown the transition of traditional RL based approaches to DRL to overcome the curse of dimensionality which is inherent in high-frequency, multi-dimensional clinical data. More recently, efforts have begun to incorporate multi-objective optimization into DRL-based personalized treatment models and are summarized in Table 5. Authors have trained their models offline on a retrospective or simulator-based data to explore the action space, and to learn an optimal value function (in value-based approaches) or an optimal policy (in policy-based approaches). From the Table 3, it is observed that the study is applied to wide area of medical field to personalize the treatment options through: suitable drug recommendation, treatment scheduling, drug administration, drug dosage recommendation, choosing a suitable combination of antibiotics, etc. Table 3 reveals that, the research work is prominently applied for critical conditions like sepsis management in ICU setting and different types of oncology. The availability of publicly available MIMIC III/IV dataset may be one of the reason for more focus in this clinical domain. The other fields touched by the researchers are: chronic conditions (Asthma, Nephrology, Rheumatoid arthritis), Neurological and degenerative disorders (anxiety, Parkinson's disease), acute surgical care (post-operative pain management, ventilator settings for COVID 19 patients). Additional domains investigated are: foot ulcer in diabetic patients, coronary artery disease, Hepatocellular carcinoma, Helicobacter pylori infection, Osteoarthritis, anxiety disorder, etc. The unavailability of publicly available high-dimensional and longitudinal dataset for specialized fields significantly hinders the development of robust DRL models for personalized treatment.

From Fig. 8, it is evident that value-based DQN and policy based PPO algorithms are explored extensively, may be due to the simplicity and availability of implementation

**Table 5** Multi-objective and DRL-based personalized treatment studies with action space

| Year | MOO + DRL technique | Objectives | Clinical action type | OPE | Safety |
|---|---|---|---|---|---|
| [75], 2024 | Multi-objective Tree-based Reinforcement Learning (MOT-RL) | Estimate tolerant dynamic treatment regimes balancing efficacy and safety | Treatment scheduling / dose adjustment (Discrete) | × | × |
| [76], 2023 | Multi-objective Deep Reinforcement Learning (MODRL) with Multi-Indicator Experience Replay | Optimize personalized drug dosing considering multiple clinical indicators | Dose optimization (Continuous) | × | × |
| [78], 2025 | Multi-task Reinforcement Learning + Explainable AI | Personalized planning and clinical decision support for orthodontic/orthognathic treatment | Treatment planning and adjustment (Discrete) | × | × |
| [79], 2025 | Reinforcement Learning + PK-PD Models (Precision Dosing) | Optimize multi-objective therapy (givinostat in polycythemia vera) | Drug dosing in multi-objective therapy (Continuous) | × | × |

support, whereas the other advanced methods like TD3, SAC, A2C, etc. still remain unexplored. In DQN, actions are discrete values, while in PPO actions can be in continuous form. DQN or any discrete action space DRL approaches are appropriate for selecting a drug from a predefined set, recommending dosage levels as low, average, high, etc., scheduling of drug dosages, etc. However, personalized treatments aim at person-specific precise drug dosing where an accurate and continuous dosage values need to be recommended. Therefore, continuous action space policy based DRL approaches are more suitable for clinical settings to model dosage optimization tailored to individual patient responses and can learn stochastic policies which are essential in healthcare settings characterized by patient variability and uncertainty. DDPG learns a deterministic policy for each action-value which makes it more suitable for problems that needs precise and continuous action values. However, its sensitivity to potential instability must be addressed by incorporating safety constraints. On the other hand, PPO learns a stochastic policy which enables it to model uncertainty in treatment decisions and capture variability in patient trajectories. CQL is a value-based approach and is inherently conservative in nature as it provides safer policy learning by avoiding unsupported or out of distribution treatment decisions. Thus, DDPG, PPO and CQL represent distinct trade-offs between action precision in treatment optimization, uncertainty-aware decision making, and safety assurance respectively. Also, the difficulty in accurate modeling of transition probability of the patient states from partially observable longitudinal patient trajectory limits the application of model-based approaches in personalized treatment. The absence of a comparative study of these algorithms in the context of personalized treatment to draw conclusions regarding their strengths and limitations, represents a significant research gap.

The offline trained RL and its variant based models need to be assessed for their robustness to distribution shift and constrained safety risks. As summarized in Table 3, only approximately 10% of the literature explicitly addressed these issues. Robustness to distribution shift is evaluated in the studies using weighted importance sampling [12], fitted Q - evaluation [57, 61], and safety risks through expert guided action filtering [45] or policy restriction mechanisms [57]. From Table 3, it is evident that authors have used different dataset, and have focused on diverse clinical domain with different objective functions which makes direct quantitative comparison challenging. Furthermore, the issues such as interpretability, safety, and scalability remains comparatively underexplored in the surveyed papers and continue to represent significant research gap.

The personalized medicine mainly relies on the multi-omics data use to identify the unique nature of patients. But, only the authors of [14] from the surveyed literature have incorporated multi-modal and genomic information. Majority of the literature rely on longitudinal vitals and laboratory results which limits agents ability to derive a personalized medicine. RL approaches which are markovian, choose the action completely based on the present state of the patient which necessitates the accurate representation of patient state. The present body of work lacks systematic integration of Multi-omics and multi-modal clinical data to accurately derive patient-specific trajectory, thereby limiting the capability of treatment personalization.

These limitations identified through the present survey highlight the need for a robust and safety-aware multi-modal multi-objective DRL framework for personalized treatment. A conceptual framework addressing these gaps is proposed in the next section.

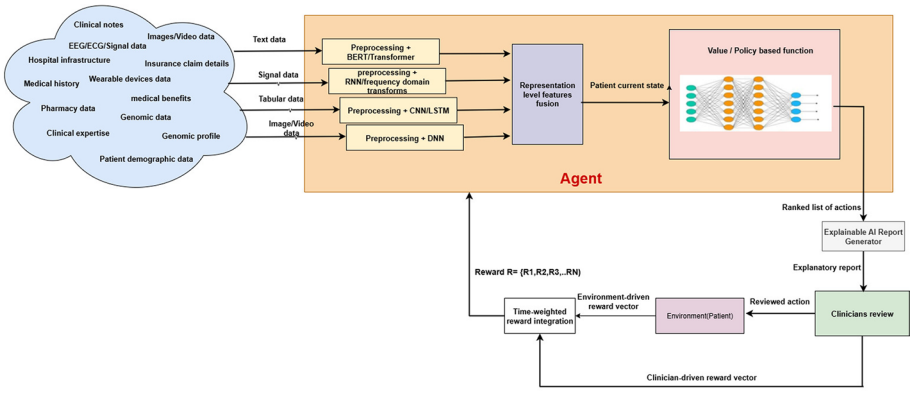



**Fig. 9** Proposed multi-modal multi-objective deep reinforcement learning

## 9 A future vision for a conceptual framework

A novel theoretical Multi-modal MODRL (MMODRL) model for personalized treatment strategy optimization is discussed here and is depicted in Fig. 9. The model is purely theoretical and has not yet been evaluated experimentally or empirically in the present study. The model is designed to integrate large-dimensional, heterogeneous patient-specific data from multiple sources and modalities including:

*Hospital information systems*: Provides access to all types of laboratory and culture reports, personal, demographic and environmental attributes, as well as longitudinal medical history from cloud-based or local servers.

*Wearable devices*: The wearable devices are connected to Internet of Things and provide continuously streaming physiological and behavioral data via wireless connectivity, irrespective of whether the patient is hospitalized, in proximity, or remotely located.

*Socioeconomic indicators*: Dynamic information on financial status of the patient which is updated regularly.

*Medical imaging*: Modalities such as MRI, X-ray, and ultrasound, which are processed through deep neural networks for feature extraction.

*Patient preferences*: Affordability plays a key role in treatment selection, as patients are more inclined toward options that are financially sustainable for them. Patient preferences need to be taken into consideration at the point of admission with a provision for subsequent updates.

*Genomic profiles*: This is an important data for personalized treatment that can be accessed from publicly available database like TCGA or ICGC, etc. and need to be linked to patient's genomic profile.

*Disease-specific biomarkers*: Molecular signatures relevant to diagnosis and prognosis.

*Insurance coverage information*: Details of the patient's active insurance plan, coverage limits, and co-payment structures.

*Hospital infrastructure availability*: Real-time information regarding bed occupancy, availability of diagnostic equipment, surgical facilities, and intensive care units.

*Hospital pricing data*: Treatment and procedure costs across hospital departments, updated dynamically to reflect resource utilization and economic constraints.

The inclusion of insurance, infrastructure, and pricing data in patient state representation is intended to ensure that the treatment strategies suggested by the MMODRL agent are not only clinically effective but also financially sustainable and logistically feasible. The modality-specific feature extractor is designed to capture relevant information

from the preprocessed multi-modal inputs. Subsequently the proposed model envisions incorporating representational feature fusion to integrate these features to represent the context-aware state of the patient. The patient state would be then fed into a DNN–driven RL agent, which is intended to recommend ranked optimized treatment strategies as actions.

Manually deriving insights from high-dimensional clinical and non-clinical patient data can be both time-consuming and extremely challenging. In this context, the MMODRL model is conceptualized as a potential decision support system that may assist in identifying and ranking treatment strategies by learning patient's state from a large number of input variables. Due to the stochastic nature of patient response to the treatment, and the presence of unseen rare conditions, etc., the MMODRL model is designed to integrate with explicit safety constraints to handle uncertainty. The policy learned by the MMODRL model when trained on the retrospective data is intended to be assessed through off-policy evaluation methods. In this context, evaluation relies on counterfactual outcome estimation instead of assuming historical clinical decisions as optimal standard. Thus, the reliability of the MMODRL is enhanced by using probabilistic calibration approaches to promote the alignment of predicted treatment strategies with empirically observed output frequencies. Also, clinician oversight would be essential to review the actions proposed by the value-based or policy-based learning agent in scenarios involving uncertainty cases, rare conditions, or out-of-distribution patient states. To facilitate transparency, and to enhance system reliability, the predicted actions can be further processed by an explainable AI (XAI) agent, which generates a human-understandable explanatory report.

The proposed actions, presented in the form of an explanatory report, would be reviewed in a practical deployment scenario by clinicians and other stakeholders, whose input serves as a form of external validation, constraint, and governance. The proposed MMODRL framework incorporates a time-weighted reward integration mechanism to account for the fact that the relative importance of treatment objectives may vary over time. This enables the framework to capture temporal changes in treatment priorities across different stages of care. In the proposed MMODRL framework (Fig. 9), the reward signal is modeled as a multi-objective vector rather than a scalar. Specifically, the reward vector is defined as $R = [R_1, R_2, R_3, ..., R_n]$, where each component represents a distinct dimension of treatment evaluation like:

- R1: Clinical outcome feedback—improvement in health indicators derived from patient monitoring systems and biomarkers.
- R2: Clinician and hospital administrative feedback—expert assessment of the appropriateness, safety, and feasibility of the suggested treatment.
- R3: Patient-reported feedback—satisfaction, quality of life, and preference alignment etc.

The objectives can be extended to financial feasibility or resource utilization efficiency as additional dimensions. The RL agent is formulated to optimize a policy over these multi-objective rewards, potentially employing techniques such as weighted scalarization pareto front optimization, or adaptive dynamic preference models. This is designed to encourage that treatment strategies are clinically effective, patient-centric, economically viable, and realistically achievable. In this way, the MMODRL agent facilitates adaptive

balancing of clinical efficacy, patient safety, financial viability, and resource availability within an iterative learning formulation. This framework proposes a closed-loop, adaptive learning system that aims to dynamically personalizes treatment strategies by harmonizing diverse medical, biological, environmental, and socioeconomic data streams.

The theoretical framework intend to inherently integrate fairness, ethical, accountability, regulatory and deployment-related aspects that are critical for real-world implementation. The inclusion of socioeconomic indicators, demographic and environmental attributes, patient preferences, insurance coverage information, and other non-clinical objectives within the proposed multi-objective structure is designed to account for contextual disparities and may contribute toward fairness-aware decision-making. Similarly, the incorporation of stakeholders—including clinicians, domain experts, and policymakers—within the feedback and validation loop is envisioned as a mechanism to support accountability, regulatory oversight, and responsible governance. By embedding human involvement and system-level actors into the conceptual learning cycle, the framework aims to acknowledge practical deployment constraints such as institutional heterogeneity and compliance requirements. While these elements are presently articulated at a theoretical level, they provide a foundation for future empirical work to formalize fairness metrics, auditing mechanisms, and deployment protocols.

## 10 Research challenges and future directions

### 10.1 Research challenges

Despite extensive research and numerous findings in the field of RL in healthcare, several critical challenges remain to be addressed.

#### *10.1.1 State representation:*

There exist significant obstacles in state representation for RL in healthcare, which involves achieving a balance between detail and computational efficiency, addressing the issue of missing data in medical records, and accurately representing important temporal and causal relationships in patient trajectories. Accurately capturing a patient's clinical state requires comprehensive, time-resolved data across multiple dimensions, which is often difficult to obtain in practice. Developing compact yet expressive state representations is critical, challenges arise due to the necessity of condensing intricate, high-dimensional, and frequently noisy medical data into more manageable formats. The absence of extensive, longitudinal data constrains the agent's capacity to precisely model patient conditions, forecast outcomes, and develop appropriate treatment strategies. Without complete records, RL systems struggle to identify temporal patterns or causal relationships, leading to biased or suboptimal decision-making. Additionally, missing data hampers the evaluation of learned policies, reducing trust and applicability in real-world clinical settings. These issues underscore the critical need for improved data integration and infrastructure development to support reliable RL deployment in diverse clinical settings.

#### *10.1.2 Integrating clinical and non-clinical objectives to reward function:*

Another major challenge lies in designing a reward function that effectively captures clinical objectives, the delayed effects and the diverse aspects of patient health complicates the alignment of RL objectives with clinical goals. The traditional survival-based

rewards may not fully reflect recovery quality, and a customized reward integrated with stakeholders feedback that prioritizes rapid stable recovery is to be used.

#### *10.1.3 Restricted action exploration and data-driven offline training:*

Assessing RL algorithms in healthcare also faces issues of patient safety, ethical concerns, and the substantial expenses linked to real-world clinical validation. The application of RL in healthcare presents considerable patient safety risks, as its trial-and-error approach could result in detrimental clinical decisions and raises ethical concerns regarding the delegation of autonomous decision-making to algorithms without clear accountability. So, it is mandatory to comprehensively train the model on the simulator. Additionally, significant financial and logistical challenges are also associated with conducting real-world clinical trials to validate RL models, which complicates widespread adoption. These factors underscore the need for cautious and well-regulated integration of RL into clinical practice.

#### *10.1.4 Non-availability of multi-dimensional and multi-modal data:*

Most of the existing studies are based on the public data. The design of an efficient and accurate personalized treatment model requires sufficient amount of data and studies with respect to each individual's case. Apart from the variables from the current healthcare system, prospective data on lifestyle, pharmacogenomic, psychosocial, molecular and biomarkers profiles, real-time data from various wearable devices, etc. is required and frequent intervention with the patients. Also, the diverse diagnostic methods, hospital infrastructure, the financial background of the patient need to be considered as the input parameters for the model.

#### *10.1.5 Institutionalization of personalized treatment strategies:*

The high cost associated with pharmacogenomic testing may slow down the process of integrating it with routine clinical trials and needs the collaboration with insurance companies and regulators, hospital administrators to come up with new regulations, required equipments, and new insurance policies. But, genomic profiling actually helps in reducing the treatment cost by recommending an optimal strategy with low drug-dose requirements and avoiding unnecessary treatments [83]. Spencer et al. [83] have reported the statistics of the cancer patients using personalized treatment options switching from immunotherapy plus chemotherapy treatment to only chemotherapy treatment which has not only resulted in cost-effectiveness, but also has improved the survival rate and the duration of survival. A lot of effort is required by all the stake holders including community workers to be effective in bringing the awareness of the outcome of personalized treatment.

### 10.2 Future directions

The future of RL in healthcare presents significant possibilities, particularly in improving individual care, combining prior knowledge, and increasing access to disadvantaged regions.

#### 10.2.1 Large language models for improving the individualized care

RL has the potential to support decision-making in healthcare by functioning as intelligent assistants in clinical decision making, enhancing the efficiency of medical documentation, improving communication with patients, and furthering medical research. LLMs can also be explored to extract information from notes written by clinicians, which is required as input for the DRL model. The well-established DRL models take the current patient status as input and output sequential treatment strategies as actions. LLMs are useful for generating a report on the customized treatment strategies decided by the DRL model. Thus, LLM models, along with DRL models, are useful in analyzing extensive patient data to deliver tailored insights, treatment recommendations, and support that align with each patient's unique medical history and requirements.

#### 10.2.2 Explainable AI for interpretable and transparent in personalized strategies

Transparency is important in clinical applications, where trust, accountability, and regulatory compliance are essential. By incorporating explainability into RL frameworks, through methods such as policy summarization, feature attribution, or interpretable models, researchers can make the agent's behavior more understandable to clinicians and stakeholders. This not only supports model validation and debugging, but also facilitates the integration of RL-based decision support systems into real-world clinical workflows, where human oversight and interpretability are paramount.

#### 10.2.3 Combining prior knowledge

Combining patient metadata, such as demographics and clinical information, with image features improves classification accuracy across various architectures, often enhancing performance metrics with little additional inference cost. Another valuable way to incorporate prior knowledge into RL for healthcare is Human-in-the-Loop (HITL). In healthcare, existing knowledge is usually seen as clinical guidelines, specialized skills, and ethical issues. HITL is being increasingly used in research and practical applications of RL for shaping the reward function, constraining unsafe actions and validating or correcting model outputs during training. HITL improves data preparation and feature engineering by incorporating clinician oversight, which enhances data quality and ensures models are ready for deployment. During the deployment phase, clinicians actively verify and adjust model outputs, including infection predictions and medication dosing recommendations. Additionally, systems can intelligently direct simpler cases to automated processes while routing more complex cases to experts, thereby optimizing efficiency while ensuring that safety is maintained. However, there are a few challenges faced in HITL learning, including concerns about data quality, such as noisy or biased annotations. It also places a burden on human experts, who are required to provide continuous feedback throughout the model training process. Furthermore, HITL systems struggle with scalability, as they must find a balance between human involvement and automation, while also ensuring accountability and transparency in AI decision-making, particularly in high-stakes fields like healthcare. Future research should focus on adaptive learning systems that judiciously solicit human input, and the advancement of ethical frameworks to govern HITL, and the development of more efficient interaction methods to reduce human burden. A standardized, domain-agnostic HITL frameworks

can facilitate broader adoption across industries and support seamless integration with next-generation AI systems.

#### *10.2.4 Increasing access to disadvantaged regions*

Rural areas, particularly in developing countries, have limited access to quality healthcare, high travel costs, and lack intelligent systems to guide hospital selection. These issues lead to delayed treatment, higher expenses, and poor health outcomes, highlighting the need for smart, data-driven recommendations. These issues can be tackled using RL based healthcare treatments. A DRL-based hospital recommendation system that optimizes decision-making by evaluating factors such as cost, distance, availability of various services, and treatment quality can improve the accessibility of healthcare in rural areas.

#### *10.2.5 Expanding the set of optimization objectives*

Research on MODRL for personalized treatment is in its early stages. Existing studies have focused primarily on a limited number of objectives, typically two to three centered on clinical outcomes. It is essential to incorporate additional objectives such as the patient's financial status, the insurance coverage, the available infrastructure at the hospital, and existing clinical expertise, etc. to recommend context- and patient preference-aware strategies. Including such non-clinical objectives will increase the relevance and practicality of the tailored treatment recommendations.

## 11 Conclusion

Multi-objective reinforcement learning provides a technically robust framework for optimizing decision-making in healthcare, particularly in contexts involving noisy, high-dimensional, and incomplete data, complex nonlinear dynamics, and sequential processes with delayed feedback. DRL has been particularly effective in personalizing healthcare, optimizing recommendations for medication, nutrition, and exercise using real-time Actor-Critic models. Architectures such as WD3QNE and tVAEs enhance sepsis management and disease progression modeling, while techniques such as DDDQN and DDQN improve treatment strategies in critical care settings. DRL further optimizes high-dimensional medical imaging and enables sophisticated intervention design using virtual body simulators. Multi-objective reinforcement learning with large language models and HITL approaches can drive safer and more transparent decision support systems. Applications also extend to cancer therapy ranking and mobile health interventions.However, persistent challenges remain, including data scarcity,learning effective state representations, defining clinically meaningful rewards, and ensuring model interpretability and safety.

To conclude, RL has the potential to support healthcare by enabling personalized, adaptive, and accessible care. The combination of RL with LLMs and HITL learning can drive safer and more transparent decision support systems, while RL-powered tools can help bridge healthcare gaps in rural and under-served regions. However, several critical hurdles must be addressed, including handling incomplete medical data, ensuring ethical and clinically aligned reward design, maintaining patient safety, and addressing privacy and scalability concerns. In the future, interdisciplinary collaboration between clinicians, data scientists, and policymakers will be the key to overcoming these challenges

and ensuring that RL technologies are deployed safely and effectively in real-world clinical practice.

**Author contributions**
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**Data availability**
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Not applicable

**Consent to participate**
Not applicable

**Consent for publication**
Not applicable

**Competing interest**
The authors declare no Competing interest.

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